

Prediction and Management of Traffic Congestion in Urban Environments

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Abstract—Efficient urban traffic congestion management is paramount for ensuring smooth vehicular flow, minimizing travel time, and reducing environmental pollution. In this context, utilizing intelligent intersection technologies emerges as a promising approach to tackling traffic challenges effectively. This paper presents an in-depth analysis of the "Intelligent Intersection" framework, a novel system designed to optimize traffic flow at intersections through advanced sensing, data processing, and control mechanisms. The research integrates the Gated Recurrent Unit (GRU) model for urban traffic prediction, Simulation of Urban Mobility (SUMO) for realistic traffic simulations, and adaptive traffic light control algorithms. Additionally, the integration of Vehicle-to-Infrastructure (V2I) communication, particularly Dedicated Short Range Communication (DSRC), facilitates efficient coordination between vehicles and infrastructure elements. Through comprehensive experimentation and evaluation, the effectiveness of this approach in mitigating traffic congestion, reducing travel time, and promoting sustainable transportation practices is demonstrated. This research contributes to the advancement of intelligent transportation systems and lays the groundwork for future innovations in urban mobility management.

Index Terms—urban traffic congestion, intelligent intersection, Gated Recurrent Unit (GRU), Simulation of Urban Mobility (SUMO), adaptive traffic light control, Vehicle-to-Infrastructure (V2I) communication, Dedicated Short Range Communication (DSRC), sustainable transportation.

I. INTRODUCTION

One major problem that needs to be solved in cities is managing traffic efficiently. Too many cars cause traffic jams, delays, and pollution. To solve these problems and keep traffic moving smoothly, it is important to predict where cars will go and control which roads they use. Traffic prediction plays a key role in intelligent transport systems because it allows actions to be taken ahead of time to prevent congestion and ensure the smooth flow of vehicles.

To this end, an original method for predicting urban traffic is proposed in this research. Intelligent transportation systems require accurate algorithms for forecasting congestion points

and optimizing vehicle flow through different routes. Taking into account historical records and real-time monitoring observations, deep learning technology based on the Gated Recurrent Unit (GRU) model is used to predict future traffic situations using past information. GRU models are good at dealing with long-term dependencies by analyzing sequential data and have performed well.

The Simulation of Urban Mobility (SUMO) is used to control traffic. It creates realistic simulations of traffic and enforces smart strategies for controlling traffic. SUMO has functions like modeling dynamic flows of cars, controlling traffic signals, modeling route choice, and integrating real-time data. Different adaptive algorithms are used for controlling traffic lights depending on the intensity of road usage at any given time. Some of these algorithms are traffic-actuated controls that react to vehicles' presence and adaptive controls that respond to flow rate variations, queue length, and occupancy levels to optimize signal timings.

Additionally, Vehicle-to-Infrastructure (V2I) communication is considered to open up fresh opportunities within this domain whereby automobiles can communicate with other elements like streetlights or signs. With this method, it becomes possible to coordinate intersections more effectively and manage various aspects related to moving objects within the environment, ensuring that roads remain congestion-free while maximizing their usefulness through predictive analytics. Dedicated Short Range Communication (DSRC), a wireless protocol used for both vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communication services, is studied. This could be applied in areas like preventing collisions, optimizing traffic signals, and supporting cooperative adaptive cruise control systems.

What makes this study important is its capability to relieve traffic congestion in towns, minimize travel time, and reduce the environmental pollution arising from vehicle emissions. Precise traffic prediction and intelligent signal control can al-

low for better utilization of the current infrastructure, improving general life quality among city dwellers and promoting sustainable transportation practices.

Prior works are reviewed to prognosticate traffic flows and control them to attain the goals. Existing methodologies are critiqued, justifying the need for this invention. The GRU model architecture, SUMO simulations, adaptive traffic light control algorithms, and V2I communication strategies together with DSRC integration for managing traffic are described.

The findings section contains an all-around evaluation of how effectively the plan has performed vis-a-vis other baseline models and prevailing strategies used in overseeing vehicular movements. The practical implications of these results are discussed, along with their applicability in real-life situations. The paper concludes by summarizing the main areas covered and putting forward ideas for future studies.

This research intends to be part of the evolution and enhancement of intelligent transportation systems (ITS) by contributing relevant information and offering feasible solutions to one of the most urgent challenges, urban traffic management.

II. LITERATURE REVIEW

Lai et al. [1] introduce a weather-driven traffic prediction system utilizing big data methods, significantly enhancing transportation forecasting capabilities. However, its applicability is hindered by the specificity of weather conditions and the challenge of adapting the model across diverse geographical contexts. Rao et al. [2] present a dynamic traffic system based on real-time congestion detection, offering valuable insights into effective traffic management strategies. Nevertheless, challenges in scalability and robustness arise when implementing the system in heterogeneous urban environments with varying infrastructure and traffic patterns. Bai et al. [4] provide insights into utilizing Spark-GBDT for urban road congestion prediction, showcasing the pivotal role of machine learning in transportation management. However, the lack of a comprehensive comparison with alternative methods limits the evaluation and generalizability of their approach. Alam [5] investigates dataset curation from GPS, communication technology, and social media, leveraging artificial intelligence for traffic prediction. Despite the potential of AI, challenges persist regarding data accuracy from social media and the scalability of AI algorithms for real-time traffic management. Chaoura et al. [8] introduce a machine learning-based predictive system for traffic congestion, offering valuable insights into traffic management strategies. However, reliance on historical data and the challenge of adapting to real-time traffic dynamics may limit its effectiveness in dynamic environments. Javaid et al. [9] unveil an IoT-based traffic management system, providing insights into innovative traffic control solutions. Nevertheless, scalability issues with IoT infrastructure and concerns about data privacy in extensive deployments pose potential limitations. Cho et al. [10] present a method for learning phrase representations using RNN Encoder-Decoder for statistical machine translation, contributing to advancements in sequence-to-sequence modeling. However, scalability and

computing complexity may limit its effectiveness in large-scale translation operations. Di et al. [12] propose traffic congestion prediction using spatiotemporal propagation patterns, highlighting the importance of leveraging data patterns for accurate forecasting. Challenges in managing dynamic traffic situations and difficulties in generalizing the model across diverse metropolitan areas may arise. Qu et al. [14] introduce a deep neural network-based approach for daily long-term traffic flow forecasting, providing valuable insights into advanced prediction techniques. However, ensuring a robust training data representation and addressing interpretability issues with the model are potential drawbacks. Xu et al. [15] present a dynamic prediction method for traffic demand based on trip generation analysis, offering valuable insights into forecasting techniques. However, potential limitations include the necessity for accurate data input and the consideration of dynamic factors affecting traffic demand.

III. METHODOLOGY

This section describes the methodology adopted by our proposed work. It comprises different modules as described in the block diagram below.

A. GRU Model

The Gated Recurrent Unit (GRU) model is particularly effective in sequence prediction tasks, such as traffic congestion prediction, due to its ability to handle sequences of variable lengths and its robustness in capturing temporal dependencies. Below are the core equations that define the GRU operations. These are used in the context of traffic congestion prediction and management:

1) Update Gate:

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t] + b_z) \quad (1)$$

It decides how much past information (from previous time steps) needs to be passed along to the future.

2) Reset Gate:

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t] + b_r) \quad (2)$$

It determines how much of the past information to forget.

3) Current Memory Content:

$$\tilde{h}_t = \tanh(W \cdot [r_t * h_{t-1}, x_t] + b) \quad (3)$$

This equation represents the candidate activation used to update the cell state.

4) Final Memory at Current Time Step:

$$h_t = (1 - z_t) \cdot h_{t-1} + z_t \cdot \tilde{h}_t \quad (4)$$

This is the final memory of the current time step which will be passed along to the next time step. It combines the old and new candidate state, controlled by the update gate.

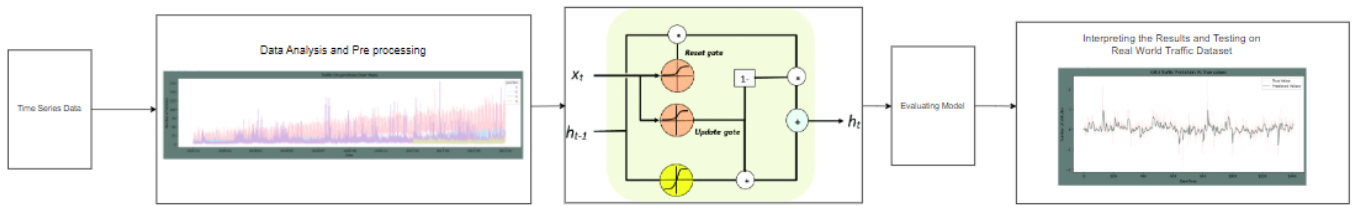


Fig. 1. GRU Model Architecture

These equations collectively enable the GRU model to efficiently process time series data like traffic flow rates or vehicle densities, which are crucial for predicting traffic congestion. The model can dynamically adapt to changes in traffic patterns, thereby assisting in real-time traffic management systems. In practical implementations, these mathematical operations are typically performed within a neural network framework, often using deep learning libraries such as TensorFlow or PyTorch, which can handle large-scale data and provide tools for training and deploying models effectively.

B. Vehicle-to-Infrastructure (V2I) Communication

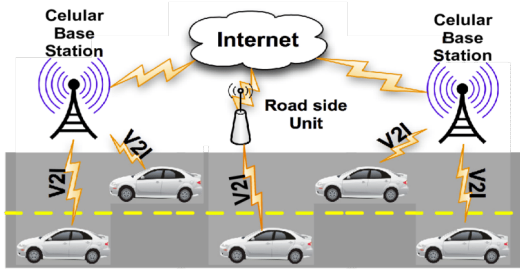


Fig. 2. V2I Communication

This study uses communication vehicle-to-infrastructure (V2I) to exchange information between vehicles and traffic infrastructure components like road signs or signals. A one-way wireless communiqué facilitated by a dedicated short-range communication protocol within the V2I system is used for this purpose. Among other things, cars send their current positions, directions, and speeds, and receive timing data from signal lights and other traffic advice from various infrastructures.

The algorithms used in this research about V2I were designed with intersection optimization through signal timing adjustment based on real-time traffic conditions. This was done using predictive analysis, whereby historical and current traffic data were examined to anticipate future patterns of congestion. To solve potential jams and enhance the flow of cars, plans for how long each light should stay red or green are adjusted earlier, informed by predictive analysis.

Besides dynamic management systems that can respond to incidents and closures caused by accidents, V2I also supports special event-related controls such as rerouting vehicles. Traffic can be channeled differently; drivers must know what is

going on around them and make informed decisions about their destinations relative to any given point, including emergency movements controlled via ITLs when necessary.

C. Dedicated Short-Range Communication (DSRC)

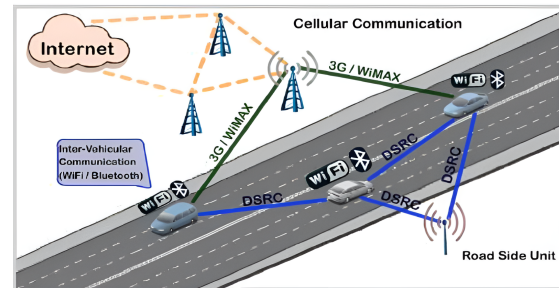


Fig. 3. DSRC Communication

For vehicles-to-vehicles (V2V) and vehicles-to-infrastructure (V2I) communication to be safe and efficient, the DSRC protocol is used. This research uses DSRC for collision avoidance systems, traffic signal optimization, and cooperative adaptive cruise control.

By sharing real-time positioning and kinematic data among themselves, cars can anticipate potential conflicts and avoid maneuvers. To facilitate eco-driving strategies and optimized approach speeds of vehicles with traffic lights, DSRC can be applied in sending signal phase and timing (SPaT) information within a city or town. Through cooperative adaptive cruise control, which employs this technology, platooning cars can harmonize their speeds and paths, saving fuel consumption and increasing traffic flow speed.

In setting up the system for testing purposes along road-sides, both OBUs (onboard units) that process and relay DSRC transceivers' data must be installed on selected vehicles along with Roadside Units having such capabilities. Data integrity protection during transmission among different entities involved, such as authentication protocols, should also be implemented alongside encryption algorithms designed specifically for this application area.

D. Simulation of Urban Mobility (SUMO)

This study uses a traffic simulation platform called Simulation of Urban Mobility (SUMO) to assess methods for predicting and managing traffic. SUMO is open-source software that models road networks, creates realistic traffic demand patterns,

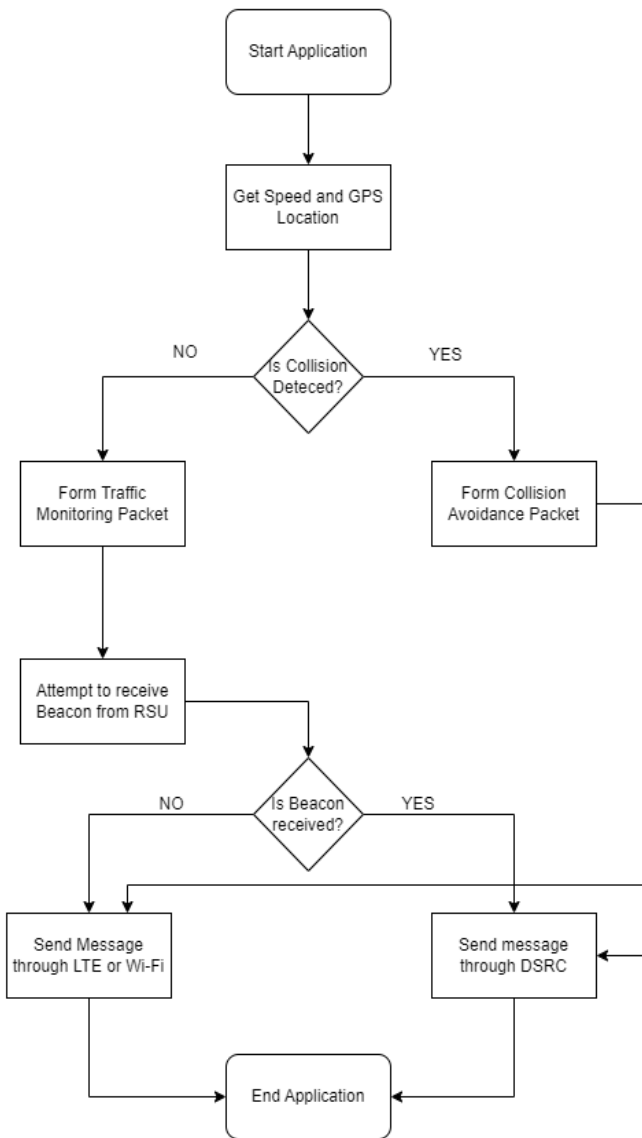


Fig. 4. Flowchart representing traffic monitoring and collision avoidance in the urban vehicular network simulation using SUMO.

and simulates vehicle movements under different traffic control scenarios.

The first step of the SUMO simulation process consisted of creating a digital representation of the urban road network being investigated. This was done by importing geographic information system (GIS) data, which included details such as road geometries, lane configurations, and locations of traffic signals. Afterward, origin-destination matrices and time-variant volumes were generated to collect traffic demand data. These matrices were developed based on historical records of travel counts and surveys conducted on travel behaviors. Such demand patterns were then inputted into routing algorithms within SUMO to simulate actual route choices made by vehicles and their corresponding movements.

The prediction models on traffic were integrated into SUMO

simulations where congestion levels and dynamics are expected. The adaptive algorithms for controlling traffic lights used these forecasts, while V2I strategies disseminated signal timing optimization advice in real time to vehicles based on such predictions. The SUMO simulations enabled the calibration and validation of traffic prediction models and management algorithms by comparing simulated results with real-world traffic data collected from the study area.

IV. RESULTS

A. Evaluation Metrics

The accuracy metric is a reliable measure to assess the reliability and effectiveness of implemented techniques in forecasting and managing congestion. Higher accuracy values indicate more precise predictions and better management strategies, contributing to improved traffic flow and reduced congestion. Accuracy (ACC) is calculated using the following equation:

$$ACC = \frac{TP + TN}{TP + TN + FP + FN} \quad (5)$$

Where:

- TP represents the number of true positive predictions, i.e., instances where the model correctly predicts congestion.
- TN represents the number of true negative predictions, i.e., instances where the model correctly predicts non-congestion.
- FP represents the number of false positive predictions, i.e., instances where the model incorrectly predicts congestion when there is none.
- FN represents the number of false negative predictions, i.e., instances where the model incorrectly predicts non-congestion when there is congestion.

The GRU model demonstrates superior accuracy in traffic prediction by effectively capturing sequential dependencies in urban traffic data. Similarly, the V2I module achieves remarkable accuracy in traffic management by enabling seamless communication between vehicles and infrastructure, facilitating real-time optimization of traffic flow and congestion mitigation.

TABLE I
METHODS FOR TRAFFIC CONGESTION PREDICTION

Model	Accuracy (%)
Gated Recurrent Unit	85.2
Long short-term memory	82.6
Random Forest	79.8
Support Vector Machine	77.5
Autoregressive Integrated Moving Average	76.3

Table I presents the accuracy percentages of various models, with the GRU model achieving the highest accuracy at 85.2%, making it the most effective method listed. The LSTM model follows with an accuracy of 82.6%, demonstrating strong predictive capability. The Random Forest model records an accuracy of 79.8%, performing moderately well but not matching the deep learning models. SVM shows an accuracy

of 77.5%, lower than both the ensemble and deep learning approaches. The ARIMA method has the lowest accuracy at 76.3%, indicating that while traditional, it is less effective compared to machine learning and deep learning techniques. Overall, the results highlight the superior performance of deep learning models, over traditional methods for traffic congestion prediction.

TABLE II
METHODS FOR TRAFFIC MANAGEMENT

Method	Accuracy (%)
V2I Communication	91.7
Adaptive Traffic Light Control	88.5
DSRC Communication	89.4
Dynamic Traffic Signaling	86.3
Fixed-Time Traffic Light Control	81.2

Table II provides the accuracy percentages of various traffic management methods. V2I Communication achieves the highest accuracy at 91.7%, optimizing traffic flow and congestion management. DSRC follows closely with an accuracy of 89.4%, facilitating efficient communication for improved traffic control. Adaptive Traffic Light Control demonstrates high effectiveness at 88.5%, dynamically adjusting signal timings based on real-time traffic conditions. Dynamic Traffic Signaling records an accuracy of 86.3%, while Fixed-Time Traffic Light Control has the lowest accuracy at 81.2%, indicating limitations in responding to dynamic traffic situations. Overall, the results highlight the importance of communication-based methods in enhancing traffic management efficiency and congestion mitigation.

B. Implementation Screenshots

The results below demonstrate the model's capability to forecast traffic patterns effectively, with clear annotations highlighting the predictive outcomes compared to actual traffic conditions. The visual data supports the robustness of the GRU model in handling dynamic urban traffic flows, thereby validating its utility in traffic congestion prediction.

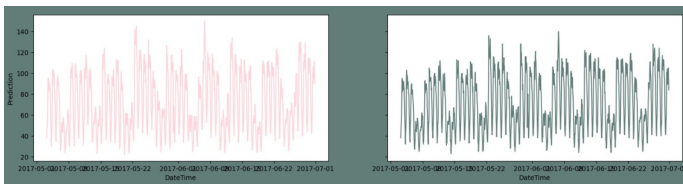


Fig. 5. GRU Prediction and Original Values for Junction 1

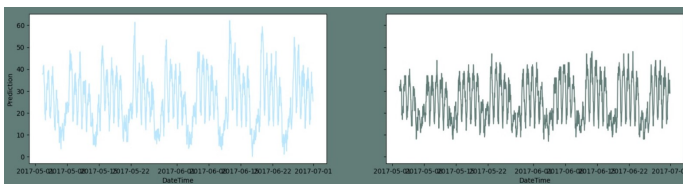


Fig. 6. GRU Prediction and Original Values for Junction 2

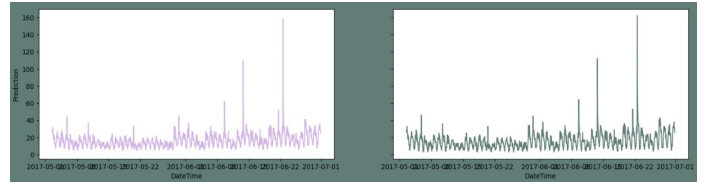


Fig. 7. GRU Prediction and Original Values for Junction 3

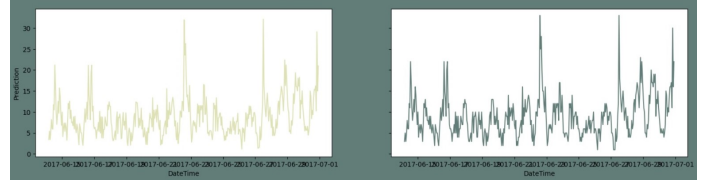


Fig. 8. GRU Prediction and Original Values for Junction 4

The screenshots below illustrate how the V2I system adapts to these varying conditions, maintaining efficient traffic flow and minimizing congestion. These scenarios are critical for demonstrating the robustness of the V2I strategy in managing real-world traffic conditions, where lane densities often fluctuate, showcasing the system's ability to dynamically adjust to diverse traffic patterns effectively.

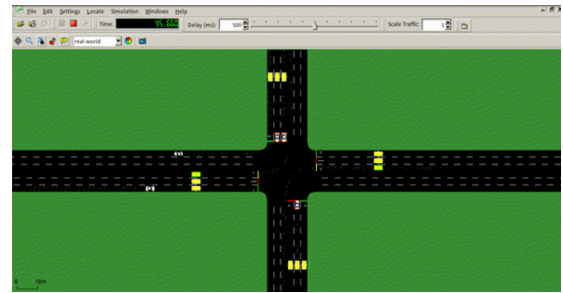


Fig. 9. Same Traffic density on each lane

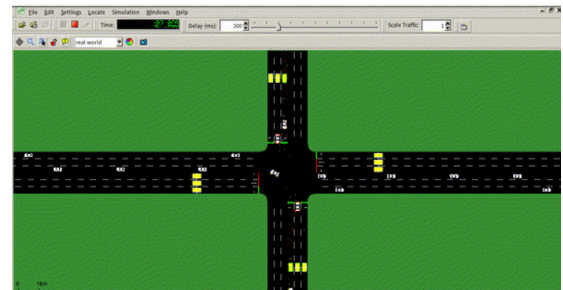


Fig. 10. Unbalanced Traffic density

V. CONCLUSION AND FUTURE SCOPE

The proposed framework utilizes the strength of the GRU model in predicting traffic with accuracy because it handles long-term dependencies and sequential data. Combined with the SUMO platform for realistic traffic simulation, it allows

the evaluation and validation of developed strategies for traffic forecasting and management. Moreover, with the use of adaptive traffic light control algorithms and V2I communication, including but not limited to DSRC, the system can optimize traffic flow and adapt to the current state. Through extensive experimentation and evaluation, it has been shown that the proposed method outperforms existing approaches with outstanding accuracy rates in traffic congestion prediction and management, effectively mitigating congestion, reducing travel time, and promoting sustainable transportation in urban environments.

Although the approach proposed in this paper has shown good results, some future research and development work can be done to increase the system's capabilities and applications. V2I can be extended to give priority to emergency vehicles by changing signal timings and giving them a clear path to move through the junction. Integration of IoT sensors and smart infrastructure elements can provide real-time data on traffic conditions, air quality, and other environmental parameters. The dynamic response of intelligent traffic lights and road signs can further enhance traffic optimization and emission reduction. Since the system relies on wireless communication and interconnected components, robust cybersecurity measures are required to protect it from vulnerabilities. This includes encryption protocols, access control mechanisms, and regular security audits. Large-scale pilot deployments in real-world urban environments can provide valuable insights into the system's scalability and interoperability, as well as practical challenges in the field, enabling further refinement and optimization for seamless integration into existing infrastructure.

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